

all $\mu \in \text{Good}$. $\mathcal{B}$ outputs 1(0) if $\mathcal{A}$ outputs 1(0) for all $\mu \in \text{Good}$. As we have explained above, $\mathcal{A}$ outputs 1(0) for all $\mu \in \text{Good}$ with high probability if $x_1, \dots, x_t$ are sampled from $U_n(\text{Gen})$, and therefore, $\mathcal{B}$ breaks the security of Gen .

However, one issue in the construction is that we do not know how to obtain Good in QPT. We therefore define another set Good' that is the set of all $\mu \in [n]$ such that $\Pr_{(y_1, \dots, y_t) \leftarrow U_n^{\otimes t}}[1 \leftarrow \mathcal{A}((\mu, y_1), \dots, (\mu, y_t))]$ is large. Such a set can be obtained in QPT, and therefore $\mathcal{B}$ can find it by itself. It is easy to verify that $\text{Good} \subseteq \text{Good}'$. Therefore, if we modify $\mathcal{B}$ as follows, $\mathcal{B}$ can correctly distinguish Gen and U_n : $\mathcal{B}$ first computes the set Good' . On input $x_1, \dots, x_t$ independently sampled from $\text{Gen}(\mu^*)$ or U_n , $\mathcal{B}$ runs $\mathcal{A}((\mu, x_1), \dots, (\mu, x_t))$ for all $\mu \in \text{Good}'$. $\mathcal{B}$ outputs 1 if $\mathcal{A}$ outputs 1 for all $\mu \in \text{Good}'$. $\mathcal{B}$ outputs 0 if $\mathcal{A}$ outputs 0 for at least one $\mu \in \text{Good}'$.

Proof of Theorem 1.3. Let us next explain our proof of Theorem 1.3, which is the equivalence between $\text{BQP} \neq \text{PP}$ and hardness of agnostic quantum distribution learning with respect to KL divergence. Our key idea is to introduce the worst-case hardness of quantum maximum likelihood (QML) estimation as an intermediate object. Let $\mathcal{D}$ be a QPT algorithm that, on input a bit string z , outputs a bit string x . Worst-case hardness of QML means that there exists a bit string x such that no QPT learner given x can find z^* such that $\Pr[x \leftarrow \mathcal{D}(z^*)]$ is close to $\max_z \Pr[x \leftarrow \mathcal{D}(z)]$. Intuitively, the worst-case hardness of QML is equivalent to the hardness of agnostic quantum distribution learning, but we focus on the former, because it is more convenient to show the equivalence to $\text{BQP} \neq \text{PP}$.⁵

Hence our goal is to show the equivalence between the worst-case hardness of QML and $\text{BQP} \neq \text{PP}$. [HH24] already showed $\text{BQP} \neq \text{PP}$ from the worst-case hardness of QML. Therefore the remaining task is to show the other direction.

Let $\mathcal{L}$ be a language in $\text{PP} = \text{PostBQP}$. Then, there exists a QPT algorithm $\mathcal{M}$ such that $\Pr_{(b, b^*) \leftarrow \mathcal{M}(x)}[b = 1 | b^* = 1] \geq \frac{2}{3}$ for all $x \in \mathcal{L}$, and $\Pr_{(b, b^*) \leftarrow \mathcal{M}(x)}[b = 1 | b^* = 1] \leq \frac{1}{3}$ for all $x \notin \mathcal{L}$. For the QPT algorithm $\mathcal{M}$, we consider $\mathcal{M}^*(x, c)$ that runs $(b, b^*) \leftarrow \mathcal{M}(x)$, and outputs x if $b = c$ and $b^* = 1$, and outputs $\perp$ otherwise. Crucially, if $x \in \mathcal{L}$, then $\Pr[x \leftarrow \mathcal{M}^*(x, 1)]$ is much larger than $\Pr[x \leftarrow \mathcal{M}^*(x, 0)]$ while if $x \notin \mathcal{L}$, then $\Pr[x \leftarrow \mathcal{M}^*(x, 0)]$ is much larger than $\Pr[x \leftarrow \mathcal{M}^*(x, 1)]$. Therefore, if QML is easy in the worst-case setting, we can check if $x \in \mathcal{L}$ or $x \notin \mathcal{L}$ in QPT by doing the maximum likelihood estimation of $\mathcal{M}^*$.

Proof of Theorem 1.11 Theorem 1.11 is the construction of a PPT algorithm with access to a Σ_3^{P} oracle that can achieve agnostic classical distribution learning with respect to the statistical distance. Let $\mathcal{T}$ be an unknown algorithm that outputs classical bit strings, and let $\mathcal{D}$ be a known QPT algorithm that takes an n -bit string z as input, and outputs a classical bit string x . The learner receives $x_1, \dots, x_t$ independently sampled from $\mathcal{T}$, and has to find z^* such that $\text{SD}(\mathcal{D}(z^*), \mathcal{T})$ is small.

In order to make our explanation simpler, let us first assume that $\mathcal{D}$'s input z is just a single bit. In this setting, access to an NP oracle, not a Σ_3^{P} oracle, is enough. Given $x_1, \dots, x_t$ independently sampled from $\mathcal{T}$, the learner has to find $b \in \{0, 1\}$ such that $\mathcal{T}$ is statistically closer to $\mathcal{D}(b)$ than $\mathcal{D}(b \oplus 1)$. How can we do that? By the definition of the statistical distance, there exists an unbounded algorithm Dis such that

$$|\Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{D}(1)] - \Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{D}(0)]| = \text{SD}(\mathcal{D}(0), \mathcal{D}(1)). \quad (2)$$

Given $x_1, \dots, x_t$ as input, which is independently sampled from $\mathcal{T}$, the learner first estimates the value of $\Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{T}]$. It then outputs 0 (1) if the estimated value of $\Pr[1 \leftarrow \text{Dis}(x) : x \leftarrow \mathcal{T}]$ is closer to

⁵Although they are equivalent at the intuition level, showing the equivalence is non-trivial, and we carefully show their equivalence, which is also our technical contribution.